

$$y'' - 3y' + 2y = 0 \Leftrightarrow Y' = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} Y \text{ où } Y = \begin{pmatrix} y' \\ y \end{pmatrix}$$

$$\Leftrightarrow Y = Y_0 e^{\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} t} \quad Y_0 = \begin{pmatrix} y'(0) \\ y(0) \end{pmatrix}$$

$$e^{\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} t} = \sum_{k=0}^{\infty} \frac{\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}^k t^k}{k!}$$

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

$$P_A(\lambda) = \begin{vmatrix} 3-\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 3\lambda + 2$$

$$P_A(\lambda) = 0 \Leftrightarrow \lambda = 1 \text{ ou } \lambda = 2$$

$$E_1 = ? \text{ Soit } v = (x, y) \in \mathbb{R}^2. \quad v \in E_1 \Leftrightarrow \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow x = y$$

$$\text{Donc } E_1 = \langle (1, 1) \rangle$$

$$E_2 = ? \text{ Soit } v = (x, y) \in \mathbb{R}^2. \quad v \in E_2 \Leftrightarrow \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow x = 2y$$

$$\text{Donc } E_2 = \langle (2, 1) \rangle$$

Il vient que

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-2}$$

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$\forall k \in \mathbb{N}$,

$$A^k = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^k \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 1 & 2^{k+2} \\ 1 & 2^k \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$$

$$A^k = \begin{pmatrix} 1+2^{k+2} & 2^{k+2} \\ 2^{k+2} & 2^{k+2} \end{pmatrix}$$

After $e^{2t} =$

$$\sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} 2e^{kt} & 2e^{kt} \\ 2^{k-1} & 2-2^k \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} (2^{k+1} - 2) & \sum_{k=0}^{\infty} \frac{1}{k!} (2-2^{k+1}) \\ \sum_{k=0}^{\infty} \frac{1}{k!} (2^k - 2) & \sum_{k=0}^{\infty} \frac{1}{k!} (2-2^k) \end{pmatrix}$$

$$= \begin{pmatrix} 2e^{2t} - e^t & 2e^t - 2e^{2t} \\ e^{2t} - e^t & 2e^t - e^{2t} \end{pmatrix}$$

Don't $Y(t) \begin{pmatrix} y'(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} y_0 \\ y_0' \\ y_0 \end{pmatrix} \begin{pmatrix} 2e^{2t} - e^t & 2e^t - 2e^{2t} \\ e^{2t} - e^t & 2e^t - e^{2t} \end{pmatrix} \begin{pmatrix} y_0'(0) \\ y_0(0) \end{pmatrix}$

$$= \begin{pmatrix} e^{2t}(2y_0'(0) - 2y_0) + e^t(y_0' + 2y_0) \\ e^{2t}(y_0' - y_0) + e^t(-y_0' + 2y_0) \end{pmatrix}$$

Don't $y(t) = \underbrace{(y_0' - y_0)}_A e^{2t} + e^t \underbrace{(-y_0' + 2y_0)}_B$